



CHAPTER

12

Atoms

Sr. No.	Concept	Formulas	Other information
1.	Distance of closest approach	K.E. of α - particle, $K = \frac{1}{2}mv^2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{2Ze^2}{r_0}$ Distance of closest approach, $r_0 = \frac{1}{4\pi\epsilon_0} \cdot \frac{2Ze^2}{K} = \frac{1}{4\pi\epsilon_0} \cdot \frac{4Ze^2}{mv^2}$	m = mass of α particle Z = atomic number e = charge of electron v = velocity of particle ϵ_0 = permittivity of free space
2.	Bohr's atomic model	Angular momentum of an electron; $L = mvr = \frac{nh}{2\pi}, \quad n = 1, 2, 3, \dots$ When electron jumps from higher energy level to lower energy level: $h\nu = E_2 - E_1$	n = principal quantum number ν = frequency of emitted radiation. h = Planck's constant E_1, E_2 = energies associated with the orbits of principal quantum numbers n_1, n_2, respectively ($n_2 > n_1$).
3.	Bohr's theory of hydrogen atom	Radius of n^{th} orbit, $r_n = \frac{n^2 h^2}{4\pi^2 m k e^2} = r_0 n^2$ $r_0 = 0.53 \text{ \AA}$ Speed of electron in n^{th} orbit, $v = \frac{2\pi k e^2}{nh} = \alpha \frac{c}{n} = \frac{1}{137} \cdot \frac{c}{n}$ Total energy of an electron in n^{th} orbit is $E_n = K.E. + P.E. = \frac{ke^2}{2r} - \frac{ke^2}{r} = -\frac{ke^2}{2r}$ or $E_n = -\frac{2\pi^2 m k^2 e^4}{n^2 h^2}$ $= \frac{-2.17 \times 10^{-18} Z^2}{n^2} \text{ J} = -\frac{13.6}{n^2} \text{ eV}$ Frequency of electron; $\nu_0 = \frac{1}{T} = \frac{v}{2\pi r} = \frac{4\pi^2 k^2 m e^4}{n^3 h^3}$ $\therefore \nu_0 \propto \frac{1}{n^3}$	R = Rydberg's constant $= 1.0973 \times 10^7 \text{ m}^{-1}$ h = Planck's constant m = mass of electron $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$ e = charge of electron $(1.6 \times 10^{-19} \text{ C})$

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4.	For hydrogen like atom (He^+ , Li^{++} etc)	$r_n = \frac{r_0 n^2}{Z}$ $v_n = \frac{Z}{n} \left(\frac{c}{137} \right)$ $E_n = \frac{-13.6 Z^2}{n^2} \text{eV}$	r_n = radius of n^{th} orbit v_n = speed of electron in n^{th} orbit E_n = energy of electron in n^{th} orbit $c = 3 \times 10^8$ m/s Z = atomic number of element of hydrogen like atom.
5.	Electron energy in hydrogen atom	$E_n = \frac{-13.6}{n^2} \text{eV}$ Relation between $K.E$ and $P.E$; $K.E = -E_n$ $P.E = 2E_n$ or $P.E = -2K.E$	E_n = energy of the n^{th} orbit. n = orbit number $K.E$ = kinetic energy of electron in n^{th} orbit. $P.E$ = potential energy of electron in n^{th} orbit.
6.	Spectral series of hydrogen atom	Frequency, $\nu = \frac{me^4}{8\epsilon_0^2 h^3} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$ Wave number, $\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$ $R = \frac{me^4}{8\epsilon_0^2 h^3 c} = 1.097 \times 10^7 \text{ m}^{-1}$	ν = frequency of electron n_1 = principal quantum number of lower energy orbit. λ = wavelength of emitted spectral line. n_2 = principal quantum number of higher energy orbit.
7.	Different spectral series for hydrogen atom	(i) Lyman Series: Here $n_2 = 2, 3, 4, \dots$ and $n_1 = 1$. $\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{1^2} - \frac{1}{n_2^2} \right]$ (ii) Balmer Series. Here $n_2 = 3, 4, 5, \dots$ and $n_1 = 2$. $\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{2^2} - \frac{1}{n_2^2} \right]$ (iii) Paschen Series. Here $n_2 = 4, 5, 6, \dots$ and $n_1 = 3$. $\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{3^2} - \frac{1}{n_2^2} \right]$ (iv) Brackett Series. Here $n_2 = 5, 6, 7, \dots$ and $n_1 = 4$. $\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{4^2} - \frac{1}{n_2^2} \right]$ (v) Pfund Series. Here $n_2 = 6, 7, 8, \dots$ and $n_1 = 5$. $\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{5^2} - \frac{1}{n_2^2} \right]$	$\bar{\nu}$ = wave number R = Rydberg's constant.